
Formatting Instructions For NeurIPS 2025

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Abstract

1

2 1 Introduction

3 In the high stakes world of algorithmic decision making, the outcomes of machine learning models
4 can have significant impacts on individuals and society in areas such as healthcare, criminal justice,
5 lending, hiring. In this contexts, fairness, is not just a desirable property but a critical requirement
6 TODO: [ADD CITATIONS]. In particular, group fairness, which requires predictive behaviour
7 remains equitable across demographically defined groups, has emerged as a key focus of research in
8 this area.

9 Notwithstanding the significant progress in developing group fairness algorithms, most of the existing
10 work assumes a stationary data distribution which, in the real world, rarely holds. Instead, data
11 distributions evolve over time: policy shifts, seasonal effects and changes in societal norms can lead
12 to distribution shifts (often referred to as concept drift) that can undermine predictive performance
13 and fairness guarantees.

14 While online learning algorithms have been created to preserve predictive accuracy, there has been
15 less work on how to preserve fairness under concept drift. This is a critical gap, because fairness
16 violations may remain undetected longer than accuracy degradation, disrupting the trustworthiness of
17 machine learning systems and leading to significant harms.

18 To address this challenge it is important not only to detect distributional changes as they occur but, at
19 the same time, characterize their impact on fairness and react accordingly. In this paper, we study
20 fairness-aware learning under concept drift in online settings and investigate how a novel framework
21 for detecting and responding to concept drift can be used to preserve fairness guarantees over time.

22 References

23 References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level
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25 is permissible to reduce the font size to small (9 point) when listing the references. Note that the
26 Reference section does not count towards the page limit.

27 [1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In
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31 *GENeral NEural Simulation System*. New York: TELOS/Springer-Verlag.

32 [3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent
33 synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

34 **A Technical Appendices and Supplementary Material**

35 Technical appendices with additional results, figures, graphs and proofs may be submitted with
36 the paper submission before the full submission deadline (see above), or as a separate PDF in the
37 ZIP file below before the supplementary material deadline. There is no page limit for the technical
38 appendices.

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- 47 • You should answer [Yes], [No], or [NA].
- 48 • [NA] means either that the question is Not Applicable for that particular paper or the
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- 50 • Please provide a short (1–2 sentence) justification right after your answer (even for NA).

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54 with the paper.

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59 "[No]" or "[NA]" is not grounds for rejection. While the questions are phrased in a binary way, we
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62 supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification
63 please point to the section(s) where related material for the question can be found.

64 IMPORTANT, please:

- 65 • **Delete this instruction block, but keep the section heading “NeurIPS Paper Checklist”,**
- 66 • **Keep the checklist subsection headings, questions/answers and guidelines below.**
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69 Question: Do the main claims made in the abstract and introduction accurately reflect the
70 paper’s contributions and scope?

71 Answer: [TODO]

72 Justification: [TODO]

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- 76 • The abstract and/or introduction should clearly state the claims made, including the
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84 Question: Does the paper discuss the limitations of the work performed by the authors?

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